INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

FINANCIAL MATHEMATICS PROJECT REPORT

Simulation and Comparative Analysis of Continuous-Time Stochastic Processes

ABSTRACT

Continuous-time stochastic differential equations (SDEs) form the core mathematical language of modern quantitative finance, yet different financial variables require markedly different drift and diffusion structures. In this project, we numerically simulate and compare four foundational continuous-time processes over a one-year trading horizon (T = 1.0 year, N = 252 daily steps, M = 10,000 Monte Carlo paths): Standard and Arithmetic Brownian Motion (BM), Geometric Brownian Motion (GBM), the Ornstein–Uhlenbeck (OU) mean-reverting process, and the Cox–Ingersoll–Ross (CIR / VIR) square-root diffusion model. We derive the analytical transition density and conditional moments for each process, implement exact transition updates for BM, GBM, and OU alongside a full-truncation Milstein discretization for the CIR model, and benchmark simulated terminal distributions against closed-form theoretical densities.

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1. Introduction

When modelling financial time series in continuous time, a central modelling question is how to specify the deterministic drift and random diffusion terms of a stochastic differential equation so that simulated trajectories respect the economic behaviour of the underlying asset or rate. The simplest continuous-time building block is the standard Wiener process (or Brownian motion), in which increments over disjoint intervals are independent and normally distributed with variance proportional to elapsed time. While arithmetic Brownian motion provides a convenient representation for cumulative log-returns or high-frequency intraday mid-price fluctuations, two well-known limitations arise when it is applied directly to longer-horizon financial variables.

First, because an arithmetic Brownian motion has constant diffusion across the entire real line, its value can become negative with positive probability, making it unsuitable for nominal stock prices under limited liability. Geometric Brownian Motion (GBM) overcomes this issue by scaling both the drift and diffusion proportionally to the current asset price S_t. As used in the Black–Scholes–Merton framework, GBM guarantees strictly positive prices and implies that continuously compounded returns follow a Gaussian distribution while price levels follow a right-skewed lognormal law.

Second, neither arithmetic nor geometric Brownian motion is stationary: the cross-sectional variance of both processes grows without bound as the time horizon t increases. While unbounded variance growth is reasonable for equity prices, it contradicts the empirical behaviour of short-term interest rates, credit spreads, and volatility indices, which tend to fluctuate within a bounded range and revert toward long-run macroeconomic averages. To capture mean reversion, the Ornstein–Uhlenbeck (OU) process—introduced to interest-rate modelling by Vasicek (1977)—adds a linear restoring drift that pulls the process back toward an equilibrium level μ. Finally, because the constant-volatility OU/Vasicek process can still wander into negative territory when volatility is high, the Cox–Ingersoll–Ross (CIR, 1985) model modifies the diffusion coefficient to scale with the square root of the current rate, σ√r_t. Alongside the Vasicek Interest Rate (VIR) specification, the CIR square-root diffusion ensures that volatility shrinks to zero as the short rate approaches the origin, keeping interest rates non-negative.

2. Mathematical Formulation of the Stochastic Models

2.1 Standard and Arithmetic Brownian Motion (BM)

A standard Brownian motion (Wiener process) {W_t}_{t>=0} satisfies three defining properties: (i) W_0 = 0 almost surely, (ii) sample paths t -> W_t are continuous with probability one, and (iii) for any 0 <= s < t, the increment W_t - W_s is independent of the past and follows a normal distribution N(0, t - s). An Arithmetic Brownian Motion X_t with constant drift rate μ and volatility σ > 0 is governed by the stochastic differential equation:

$$dX_t = \mu\, dt + \sigma\, dW_t, \quad X_0 = x_0. \quad (1)$$

Integrating Equation (1) from 0 to t yields $X_t = X_0 + \mu t + \sigma W_t$. Because $W_t \sim N(0, t)$, the state variable at any horizon t is normally distributed with conditional mean $E[X_t] = X_0 + \mu t$ and conditional variance $Var(X_t) = \sigma^2 t$.

2.2 Geometric Brownian Motion (GBM)

Under Geometric Brownian Motion, the percentage return dS_t / S_t is modelled as an arithmetic Brownian motion. For an initial asset price $S_0 > 0$, expected return μ, and percentage volatility $\sigma > 0$, the dynamics follow:

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (2)$$

Applying Itô's Lemma to $f(S_t) = \ln S_t$ introduces the quadratic variation convexity correction $-\frac{1}{2}\sigma^2 dt$, yielding the exact closed-form trajectory:

$$S_t = S_0 \exp((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t). \quad (3)$$

Consequently, $\ln(S_t / S_0) \sim N((\mu - \frac{1}{2}\sigma^2)t, \sigma^2 t)$. The expected asset price is $E[S_t] = S_0 \exp(\mu t)$, the median price is $Med[S_t] = S_0 \exp((\mu - \frac{1}{2}\sigma^2)t)$, and the conditional variance is $Var(S_t) = S_0^2 \exp(2\mu t)(\exp(\sigma^2 t) - 1)$.

2.3 The Ornstein–Uhlenbeck (OU) Process

The Ornstein–Uhlenbeck process models a mean-reverting variable X_t that is pulled toward a long-run equilibrium μ at speed $\theta > 0$ with constant volatility $\sigma > 0$:

$$dX_t = \theta(\mu - X_t) dt + \sigma dW_t. \quad (4)$$

Multiplying both sides by the integrating factor $\exp(\theta t)$ and integrating gives:

$$X_t = \mu + (X_0 - \mu) \exp(-\theta t) + \sigma \int_0^t \exp(-\theta(t - s)) dW_s. \quad (5)$$

By Itô isometry, X_t is Gaussian with conditional mean $E[X_t] = \mu + (X_0 - \mu)\exp(-\theta t)$ and variance $Var(X_t) = (\sigma^2 / 2\theta)(1 - \exp(-2\theta t))$. As $t \to \infty$, X_t converges to the stationary distribution $N(\mu, \sigma^2 / 2\theta)$ with half-life $t_{1/2} = \ln(2)/\theta$.

2.4 The Cox–Ingersoll–Ross (CIR / VIR) Square-Root Model

To prevent the negative interest rates permitted by the Vasicek/OU process, Cox, Ingersoll, and Ross (1985) replaced constant volatility with a square-root diffusion term $\sigma\sqrt{r_t}$:

$$dr_t = \kappa(\theta - r_t) dt + \sigma \sqrt{r_t} dW_t, \quad r_0 > 0. \quad (6)$$

When r_t approaches zero, the diffusion term vanishes while positive drift $\kappa\theta$ dt pushes the rate upward. When the Feller condition $2\kappa\theta >= \sigma^2$ holds, the zero boundary is unattainable and r_t remains strictly positive for all $t >= 0$. The exact transition law at horizon t is a scaled non-central chi-square distribution, $2c \cdot r_t \sim \chi'^2(\nu, \lambda_t)$, with scale $c = 2\kappa / [\sigma^2(1 - \exp(-\kappa t))]$, degrees of freedom $\nu = 4\kappa\theta/\sigma^2$, and non-centrality parameter $\lambda_t = 2c \cdot r_0 \exp(-\kappa t)$.

Model	SDE Specification	Transition Law	Conditional Mean	Conditional Variance	Domain
Brownian Motion (BM)	dX_t = μ dt + σ dW_t	Normal N(X_0+μt, σ²t)	X_0 + μt	σ²t	(-∞, +∞)
Geometric BM (GBM)	dS_t = μS_t dt + σS_t dW_t	Lognormal	S_0 exp(μt)	S_0² e^{2μt}(e^{σ²t}-1)	(0, +∞)
Ornstein–Uhlenbeck	dX_t = θ(μ - X_t)dt + σ dW_t	Normal (AR(1) limit)	μ + (X_0-μ)e^{-θt}	(σ²/2θ)(1 - e^{-2θt})	(-∞, +∞)
CIR / VIR Model	dr_t = κ(θ - r_t)dt + σ√r_t dW_t	Non-Central χ'²(ν, λ_t)	θ + (r_0-θ)e^{-κt}	r_0(σ²/κ)(e^{-κt}-e^{-2κt}) + θ(σ²/2κ)(1-e^{-κt})²	[0, +∞)

3. Numerical Discretization and Simulation Methodology

All four processes are simulated over a one-year horizon (T = 1.0 year) partitioned into N = 252 equal trading days (Δt = 1/252 ≈ 0.003968 years) across M = 10,000 independent Monte Carlo paths using NumPy's PCG64 generator (seed = 42). For BM, GBM, and OU, we step forward using the exact closed-form transition distributions in Equations (1), (3), and (5). For the CIR / VIR square-root diffusion, we implement the full-truncation Milstein scheme with reflection at zero, where r_k^+ = max(r_k, 0) and ΔW_k = √Δt Z_k:

$$r_{k+1} = \max(0,\ r_k + \kappa(\theta - r_k^+)\Delta t + \sigma\sqrt{r_k^+}\,\Delta W_k + \tfrac{1}{4}\sigma^2((\Delta W_k)^2 - \Delta t)). \quad (7)$$

4. Simulation Results and Discussion

4.1 Non-Stationary Price Paths: Brownian Motion vs. GBM

Figure 1 compares the simulated sample paths and terminal distributions at T = 1.0 year for Arithmetic Brownian Motion (X_0 = 0.0, μ = 0.15, σ = 0.30) and Geometric Brownian Motion (S_0 = 100.0, μ = 0.12, σ = 0.25). In Figure 1(a), the 5th–95th percentile envelope of ABM widens symmetrically around μt, and several paths cross below zero (reaching -1.45). In Figure 1(c)–(d), GBM trajectories remain strictly positive (min = 41.64) and form a right-skewed terminal lognormal distribution (skewness = 0.7515) where the sample mean (113.37) exceeds the median (109.28) due to the Itô convexity adjustment -½σ².

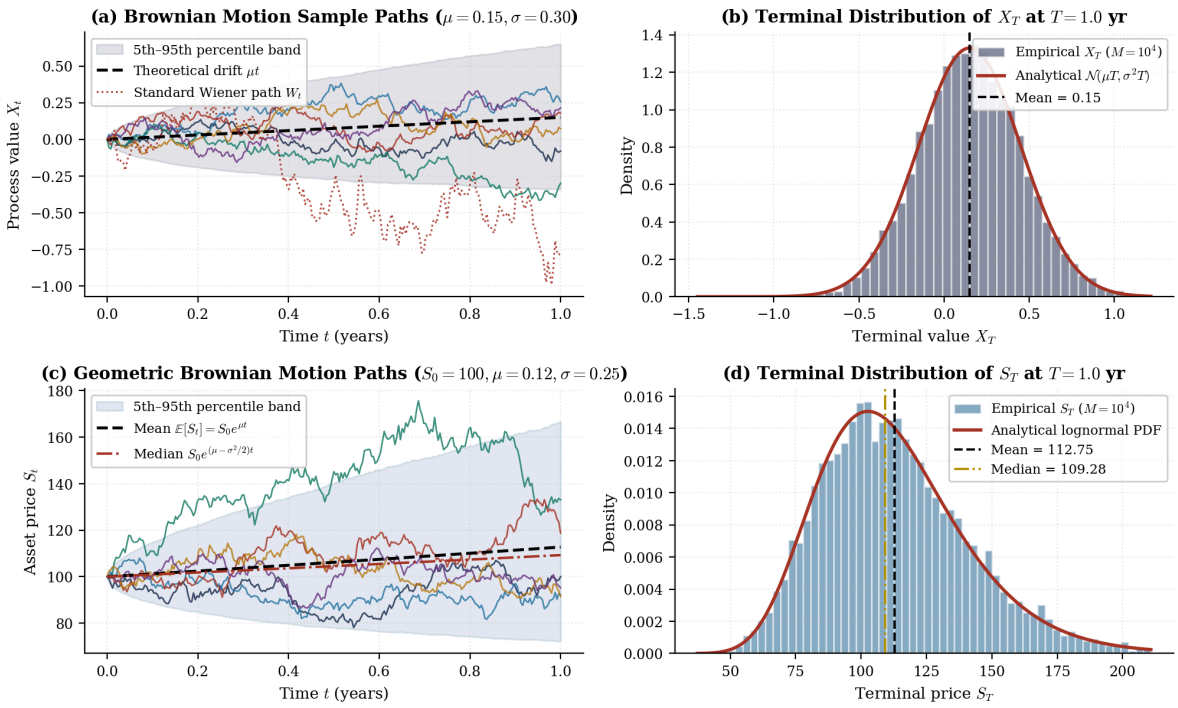


Figure 1: Monte Carlo simulation of Arithmetic Brownian Motion (top) and Geometric Brownian Motion (bottom) across M = 10,000 paths and N = 252 steps, showing sample trajectories, 5th–95th percentile bands, and terminal empirical histograms matched against analytical Gaussian and lognormal densities.

4.2 Mean-Reverting Rate Dynamics: OU vs. CIR / VIR

Figure 2 presents the simulation results for the Ornstein–Uhlenbeck process (X_0 = 10%, μ = 5%, θ = 4.0, σ = 3%) and the Cox–Ingersoll–Ross (CIR / VIR) model (r_0 = 8%, θ = 5%, κ = 3.5, σ = 8%). Both models decay exponentially from their initial states toward the 5.0% equilibrium rate within roughly 40–50 trading days, after which their 5th–95th percentile envelopes stabilize. Under the CIR parameter set, 2κθ = 0.3500 >= σ^2 = 0.0064 satisfies the Feller condition, and the terminal short-rate histogram aligns closely with the analytical non-central chi-square density.

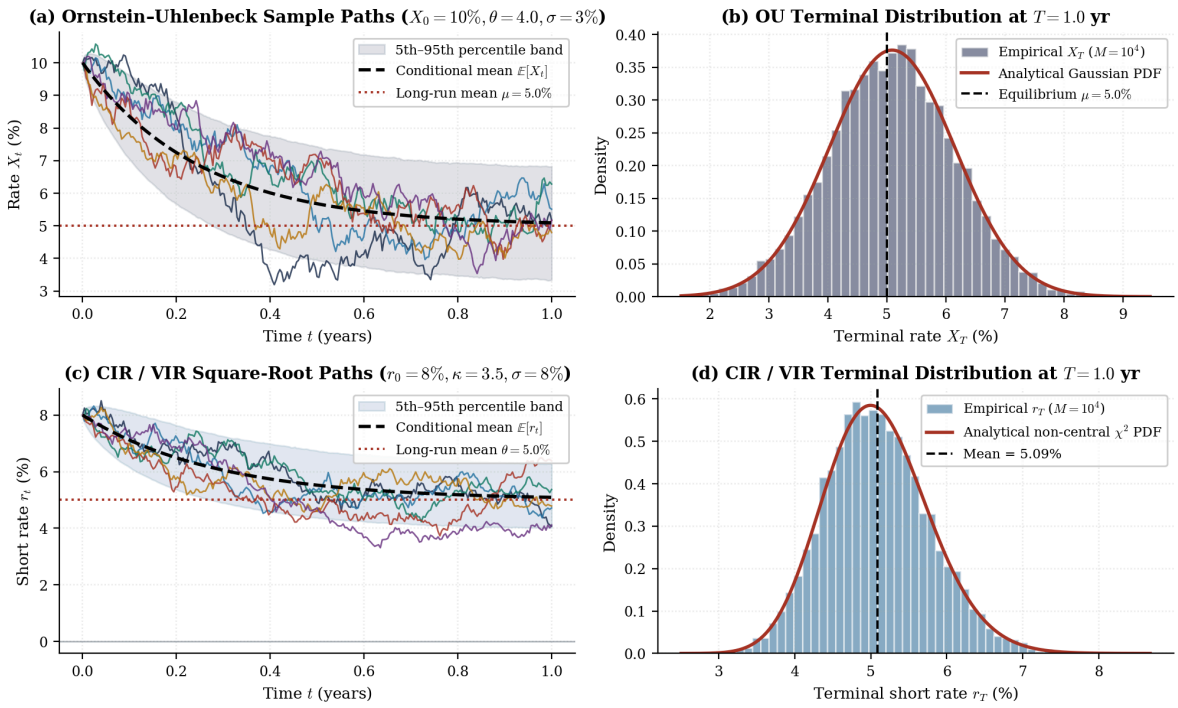


Figure 2: Monte Carlo simulation of the Ornstein–Uhlenbeck process (top) and Cox–Ingersoll–Ross (CIR / VIR) square-root diffusion (bottom) across M = 10,000 paths, comparing mean-reverting sample trajectories and terminal empirical densities against exact Gaussian and non-central χ'^2 distributions.

4.3 Moment Convergence and Zero-Boundary Behaviour

Table 2 benchmarks the empirical terminal moments across all M = 10,000 simulated paths against exact closed-form theoretical values at T = 1.0 year. Across all four models, simulated means and standard deviations match analytical expectations within one Monte Carlo standard error.

Model & Parameters	Theo. Mean	Sim. Mean	Theo. Std	Sim. Std	Skewness	Ex. Kurtosis	Simulated [Min, Max]
Brownian Motion (BM) X_0=0, μ=0.15, σ=0.30	0.1500	0.1526	0.3000	0.2999	0.0343	0.0461	[-1.4488, +1.2148]
Geometric BM (GBM) S_0=100, μ=0.12, σ=0.25	112.7497	113.3699	28.6336	29.0791	0.7515	0.9839	[41.6387, 275.2804]

Model & Parameters	Theo. Mean	Sim. Mean	Theo. Std	Sim. Std	Skewness	Ex. Kurtosis	Simulated [Min, Max]
OU Process (Vasicek) X_0=10%, μ=5%, θ=4.0, σ=3%	0.05092	0.05081	0.01061	0.01063	0.0126	-0.0405	[0.01518, 0.09462]
CIR / VIR Diffusion r_0=8%, θ=5%, κ=3.5, σ=8%	0.05091	0.05084	0.00688	0.00690	0.2884	0.1671	[0.02787, 0.08259]

Figure 3(a) illustrates how normalized variance Var(X_t)/Var(X_T) grows linearly for BM and convexly for GBM, whereas both OU and CIR saturate rapidly at their stationary ceiling. Figure 3(b) subjects the OU/Vasicek model and two CIR parameterizations to the exact same sequence of Wiener shocks starting at r_0 = 2.5%: while the additive-noise OU trajectory breaches 0% and drops to -4.66%, the CIR processes reflect at 0% or remain strictly positive under the Feller condition.

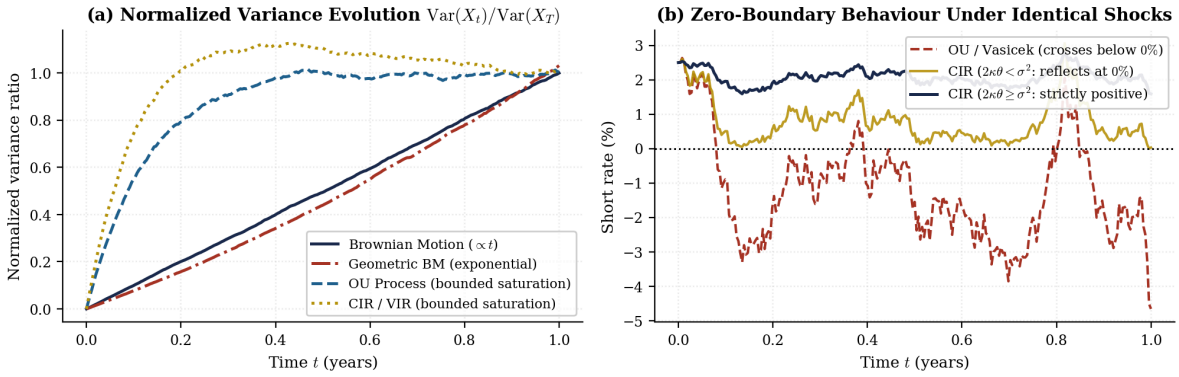


Figure 3: (a) Normalized variance evolution Var(X_t)/Var(X_T) contrasting unbounded diffusion in BM and GBM with stationary saturation in OU and CIR; (b) zero-boundary comparison on shared Wiener shocks showing negative rates under OU/Vasicek versus non-negative paths under CIR / VIR.

5. Conclusion

This simulation study demonstrates how the specification of drift and diffusion terms governs both the short-term trajectory geometry and long-run distributional behaviour of continuous-time stochastic models. Standard and Arithmetic Brownian motion provide an exact Gaussian baseline with linearly growing variance, while Geometric Brownian Motion introduces multiplicative diffusion that preserves asset-price positivity and generates a right-skewed lognormal distribution. For mean-reverting quantities such as interest rates, both the Ornstein–Uhlenbeck (Vasicek)

process and the Cox–Ingersoll–Ross (CIR / VIR) model bound long-horizon variance around a stationary equilibrium, with the square-root diffusion of the CIR model eliminating negative rate excursions under the Feller condition.

Appendix: Python Simulation Code

```
import numpy as np
from scipy import stats

def simulate_stochastic_models(T=1.0, N=252, M=10000, seed=42):
    rng = np.random.default_rng(seed)
    dt = T / N
    t = np.linspace(0.0, T, N + 1)

    # 1. Arithmetic Brownian Motion (BM): dX_t = mu*dt + sigma*dW_t
    bm_mu, bm_sigma, bm_x0 = 0.15, 0.30, 0.0
    dW_bm = rng.normal(0.0, np.sqrt(dt), size=(M, N))
    W_bm = np.hstack([np.zeros((M, 1)), np.cumsum(dW_bm, axis=1)])
    X_bm = bm_x0 + bm_mu * t + bm_sigma * W_bm

    # 2. Geometric Brownian Motion (GBM): dS_t = mu*S_t*dt + sigma*S_t*dW_t
    gbm_s0, gbm_mu, gbm_sigma = 100.0, 0.12, 0.25
    dW_gbm = rng.normal(0.0, np.sqrt(dt), size=(M, N))
    W_gbm = np.hstack([np.zeros((M, 1)), np.cumsum(dW_gbm, axis=1)])
    S_gbm = gbm_s0 * np.exp((gbm_mu - 0.5 * gbm_sigma**2) * t + gbm_sigma *
W_gbm)

    # 3. Ornstein-Uhlenbeck (OU) Process: dX_t = theta*(mu - X_t)*dt +
sigma*dW_t
    ou_x0, ou_mu, ou_theta, ou_sigma = 0.10, 0.05, 4.0, 0.03
    X_ou = np.zeros((M, N + 1)); X_ou[:, 0] = ou_x0
    exp_th = np.exp(-ou_theta * dt)
    ou_std = ou_sigma * np.sqrt((1.0 - np.exp(-2.0 * ou_theta * dt)) / (2.0 *
ou_theta))
    Z_ou = rng.normal(0.0, 1.0, size=(M, N))
    for k in range(N):
        X_ou[:, k + 1] = ou_mu + (X_ou[:, k] - ou_mu) * exp_th + ou_std *
Z_ou[:, k]

    # 4. CIR / VIR Square-Root Model: dr_t = kappa*(theta - r_t)*dt +
sigma*sqrt(r_t)*dW_t
    cir_r0, cir_theta, cir_kappa, cir_sigma = 0.08, 0.05, 3.5, 0.08
    r_cir = np.zeros((M, N + 1)); r_cir[:, 0] = cir_r0
    dW_cir = rng.normal(0.0, np.sqrt(dt), size=(M, N))
    for k in range(N):
        r_pos = np.maximum(r_cir[:, k], 0.0)
        dr = (cir_kappa * (cir_theta - r_pos) * dt
              + cir_sigma * np.sqrt(r_pos) * dW_cir[:, k]
              + 0.25 * (cir_sigma**2) * (dW_cir[:, k]**2 - dt))
        r_cir[:, k + 1] = np.maximum(r_cir[:, k] + dr, 0.0)

    return t, X_bm, S_gbm, X_ou, r_cir
```